

On the Resolving Power of the Active Corner Camera

Sheila W. Seidel, Hoover Rueda-Chacon, Christopher Yu, and Vivek K Goyal

[I think that a title with more emphasize on comparison or the power of the edge would be a good way to differentiate from our NPHOT letter. Idea: “On the Resolving Power of Edge Occlusion in Active Non-Light-of-Sight Tracking”. (I don’t love that, just want to put propose a very different title.) –VKG][Oops! I don’t like the title I left in here either. Let’s definitely brainstorm and improve. –SS]

I. INTRODUCTION

The ability to form non-line-of-sight (NLOS) images of changing scenes could be transformative in a variety of fields, including search and rescue, autonomous vehicle navigation, and reconnaissance. Most existing active NLOS methods illuminate the hidden scene using a pulsed laser directed at a relay surface and collect time-resolved measurements of returning light [1]. The prevailing approaches include raster scanning of a rectangular grid on a vertical wall opposite the volume of interest to generate a collection of confocal measurements [2]–[5]. These and a recent method that uses a horizontal relay surface [6] are inherently limited by the need for laser scanning. Methods that avoid laser scanning track the moving parts of the hidden scene as one or two point targets [7], [8]. In [Cite our letter], the *active corner camera* was introduced as a way to form accurate reconstructions of objects in motion as well as a ‘map’ of the stationary scenery behind them. [I’m imagining this going on arXiv and not to a journal so I borrowed whole sentences from the abstract of our NP letter. Please let me know if that’s not okay to do. –SS]

The active corner camera consists of pulsed laser illumination and a single-photon avalanche diode (SPAD) array detector, both directed at a visible surface close to the occluding edge as shown in Figure 1. The edge blocks light from scene as a function of its azimuthal angle, giving rise to azimuthal resolution [6], [9]–[12]. The goal is to use the measurements acquired by the SPAD array to reconstruct the scene hidden from view around the occluding edge. The setup in [8] uses similar hardware but positions the SPAD FOV away from the occluding edge to avoid the effects of occlusion.

In this work, we derive (and develop tools to compute) the Cramér–Rao bound for estimating the parameters of a hidden target using the active corner camera setup in Figure 1. Our goal is to use these tools to provide a qualitative comparison

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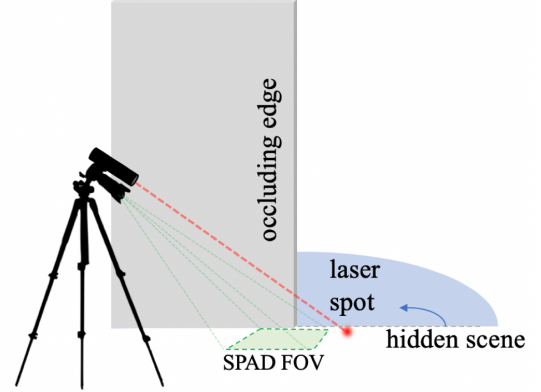


Fig. 1: In the active corner camera imaging scenario, we seek an estimate of the scene hidden from view around the corner. The hidden scene is illuminated using a pulsed laser pointed at the floor. Light returning from the hidden scene is measured using a SPAD array pointed at the floor adjacent to the occluding edge.

among different acquisition strategies. Our key contributions include:

- A demonstration of the merit of exploiting the occlusion provided by the vertical edge (as in [Cite our letter]) over the approach of avoiding the effects of occlusion (as in [8]).
- Plots of estimate uncertainty versus various system parameters, including FOV size, sensor pixel dimension and sensor time resolution. These plots provide intuition about the active corner camera and may be used to optimize the setup for different operational needs.
- A plot of the uncertainty reduction achieved when the dwell time is divided between two different laser positions.

II. CRAMÉR-RAO BOUND DERIVATION

Consider Figure 2. A pulsed laser illuminates a surface with normal vector $\hat{n}_l$ at point p_l . An infinitesimally small vertical facet at point p_s in the hidden space and with surface normal $\hat{n}_s$ reflects light. Point p_f is on the floor, with surface normal $\hat{n}_f$, in the SPAD FOV. The SPAD camera itself is located at point p_c . We parameterize the hidden scene in terms of cylindrical coordinates so that $p_s = [\rho \cos \phi, \rho \sin \phi, h]$ and $\hat{n}_s = [-\cos \phi, -\sin \phi, 0]$ where, with the origin at the base of the occluding edge, ϕ is the polar angle, ρ is the polar range, and h is the vertical height.

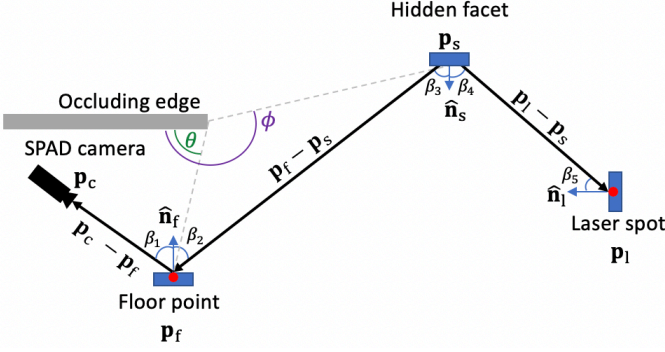


Fig. 2: A pulsed laser illuminates point $\mathbf{p}_l$. Light is reflected from an infinitesimally small facet in the hidden scene at point $\mathbf{p}_s$, then strikes the floor on the visible side at point $\mathbf{p}_f$, and is measured by the SPAD camera located at point $\mathbf{p}_c$.

The light outgoing at time t at point $\mathbf{p}_f$ due to the hidden facet at $\mathbf{p}_s$ is

$$L_h(\mathbf{p}_f, t) = \frac{\alpha H(\pi + \theta - \phi) f(\mathbf{p}_c, \mathbf{p}_s, \mathbf{p}_l, \mathbf{p}_f) s(t - \frac{\|\mathbf{p}_s - \mathbf{p}_l\| + \|\mathbf{p}_s - \mathbf{p}_f\|}{c})}{8\pi^3 \|\mathbf{p}_c - \mathbf{p}_f\|^2 \|\mathbf{p}_s - \mathbf{p}_l\|^2 \|\mathbf{p}_s - \mathbf{p}_f\|^2}, \quad (1)$$

where $s(\cdot)$ is the pulsed waveform, $H(x)$ is the Heaviside step function, and α is the albedo of the point target. With angles $\beta_1, \beta_2, \beta_3$, and β_4 defined in fig. 2, the function $f(\mathbf{p}_s, \mathbf{p}_l, \mathbf{p}_f)$ describes the effect of foreshortening:

$$f(\mathbf{p}_c, \mathbf{p}_s, \mathbf{p}_l, \mathbf{p}_f) = \cos(\beta_1) \cos(\beta_2) \cos(\beta_3) \cos(\beta_4) \cos(\beta_5) \\ = \frac{\hat{\mathbf{n}}_f \cdot (\mathbf{p}_c - \mathbf{p}_f)}{\|\mathbf{p}_c - \mathbf{p}_f\|} \cdot \frac{\hat{\mathbf{n}}_f \cdot (\mathbf{p}_s - \mathbf{p}_f)}{\|\mathbf{p}_f - \mathbf{p}_s\|} \cdot \frac{\hat{\mathbf{n}}_s \cdot (\mathbf{p}_f - \mathbf{p}_s)}{\|\mathbf{p}_f - \mathbf{p}_s\|} \cdot \frac{\hat{\mathbf{n}}_s \cdot (\mathbf{p}_l - \mathbf{p}_s)}{\|\mathbf{p}_l - \mathbf{p}_s\|} \cdot \frac{\hat{\mathbf{n}}_l \cdot (\mathbf{p}_s - \mathbf{p}_l)}{\|\mathbf{p}_l - \mathbf{p}_s\|} \quad (2)$$

The Poisson rate, due only to the hidden facet, for a camera measurement q , corresponding to time j and spatial location i is the integral of L_0 over each pixel in space and time:

$$\mathbf{y}_q = \mathbf{y}_{(i,j)} = \int_{(j-1)\Delta t}^{j\Delta t} \int_{\mathbf{p} \in \mathcal{P}_i} L_h(\mathbf{p}_f, t) d\mathbf{p} dt \\ = \int_{(j-1)\Delta t}^{j\Delta t} \int_{\mathbf{p} \in \mathcal{P}_i} \frac{\alpha H(\pi + \theta - \phi) f(\mathbf{p}_s, \mathbf{p}_l, \mathbf{p}_f) s(t - \frac{\|\mathbf{p}_s - \mathbf{p}_l\| + \|\mathbf{p}_s - \mathbf{p}_f\|}{c})}{8\pi^3 \|\mathbf{p}_c - \mathbf{p}_f\|^2 \|\mathbf{p}_s - \mathbf{p}_l\|^2 \|\mathbf{p}_s - \mathbf{p}_f\|^2} d\mathbf{p} dt. \quad (3)$$

We seek the Cramér–Rao bound for hidden targets parameters: $\Psi = [\alpha, \rho, \phi, h,]^T$, based on measurement $\mathbf{x} = \text{Poisson}(\mathbf{y} + \mathbf{b})$, where $\mathbf{b}$ is the background rate. The Poisson likelihood of measurement $\mathbf{x}$ given parameters Ψ is

$$p(\mathbf{x}; \Psi) = \prod_q \frac{(\mathbf{y}_q + \mathbf{b}_q)^{\mathbf{x}_q}}{\mathbf{x}_q!} \exp(-(\mathbf{y}_q + \mathbf{b}_q)). \quad (4)$$

The partial derivative with respect to an unknown parameter given by

$$\frac{\partial}{\partial \Psi_n} \log(p(\mathbf{x}; \Psi)) = \sum_{q=1}^N \left[\frac{\partial}{\partial \Psi_n} \mathbf{x}_q \log(\mathbf{y}_q + \mathbf{b}_q) - \frac{\partial}{\partial \Psi_n} (\mathbf{y}_q) \right]. \quad (5)$$

When we define the total rates at pixel q to be $g_q(\Psi) := \mathbf{y}_q + \mathbf{b}_q$. Applying the chain rule to the first term, eq. (5) may be rewritten:

$$\frac{\partial}{\partial \Psi_n} \log(p(\mathbf{x}; \Psi)) = \sum_{q=1}^N \left(\mathbf{x}_q \frac{1}{g_q(\Psi)} \frac{\partial}{\partial \Psi_m} (g_q(\Psi)) - \frac{\partial}{\partial \Psi_m} (g_q(\Psi)) \right). \quad (6)$$

The entries of the Fisher information matrix may be computed as follows

$$\mathbf{I}_{m,k} = \mathbb{E} \left[\frac{\partial}{\partial \Psi_m} \log(p(\mathbf{x}; \Psi)) \frac{\partial}{\partial \Psi_k} \log(p(\mathbf{x}; \Psi)) \right] \quad (7) \\ = \mathbb{E} \left[\sum_{q=1}^N \left(\mathbf{x}_q \frac{1}{g_q(\Psi)} \frac{\partial}{\partial \Psi_m} (g_q(\Psi)) - \frac{\partial}{\partial \Psi_m} (g_q(\Psi)) \right) \cdot \sum_{u=1}^N \left(\mathbf{x}_u \frac{1}{g_u(\Psi)} \frac{\partial}{\partial \Psi_k} (g_u(\Psi)) - \frac{\partial}{\partial \Psi_k} (g_u(\Psi)) \right) \right], \quad (8)$$

where (a) follows from a substitution of (6) into (7). The term $g_q(\Psi)$ is deterministic so $\mathbf{x}_q$ and $\mathbf{x}_u$ are the only random variables inside of the expectation. Expanding Equation (8) results in an expression that is linear in $\mathbf{x}_q$ and $\mathbf{x}_q^2$, so we compute $\mathbb{E}[\mathbf{x}_q]$ and $\mathbb{E}[\mathbf{x}_q^2]$.

The entries of $\mathbf{x}_q$ are independent from one another and Poisson distributed according to $\mathbf{x}_q \sim \text{Poisson}(\mathbf{y}_q + \mathbf{b}_q)$. Therefore, $\mathbb{E}[\mathbf{x}_q] = \mathbf{y}_q + \mathbf{b}_q = g_q(\Psi)$ and $\mathbb{E}[\mathbf{x}_q^2] = \text{Var}(\mathbf{x}_q) + \mathbb{E}[\mathbf{x}_q]^2 = (\mathbf{y}_q + \mathbf{b}_q) + (\mathbf{y}_q + \mathbf{b}_q)^2 = g_q(\Psi) + g_q^2(\Psi)$. Note that in this case, cross terms in Equation (8) (where $q \neq u$) go to zero. Accounting for remaining terms, where $q = u$, eq. (8) reduces to:

$$\mathbf{I}_{m,k} = \sum_{q=1}^N \frac{\partial}{\partial \Psi_m} (g_q(\Psi)) \cdot \frac{\partial}{\partial \Psi_k} (g_q(\Psi)) \cdot \frac{1}{g_q(\Psi)} \quad (9)$$

where

$$\frac{\partial}{\partial \Psi_m} g_q(\Psi) = \frac{\partial}{\partial \Psi_m} \mathbf{y}_q + \frac{\partial}{\partial \Psi_m} \mathbf{b}_q \\ = \frac{\partial}{\partial \Psi_m} \mathbf{y}_q \\ = \int_{(j-1)\Delta t}^{j\Delta t} \int_{\mathbf{p} \in \mathcal{P}_i} \frac{\partial}{\partial \Psi_m} \frac{\alpha H(\pi + \theta - \phi) f(\mathbf{p}_s, \mathbf{p}_l, \mathbf{p}_f) s(t - \frac{\|\mathbf{p}_s - \mathbf{p}_l\| + \|\mathbf{p}_s - \mathbf{p}_f\|}{c})}{8\pi^3 \|\mathbf{p}_c - \mathbf{p}_f\|^2 \|\mathbf{p}_s - \mathbf{p}_l\|^2 \|\mathbf{p}_s - \mathbf{p}_f\|^2} d\mathbf{p} dt. \quad (10)$$

The Cramér–Rao bound on the unknown parameters may be computed in term of the Fisher information matrix

$$\text{CRB}(\Psi_m) = [\mathbf{I}^{-1}]_{m,m}. \quad (11)$$

The derivative inside of the integrand in (10) was computed for each Ψ_m using Wolfram Mathematica and exported to

MATLAB code. The integrals in (10) were computed using numerical integration.

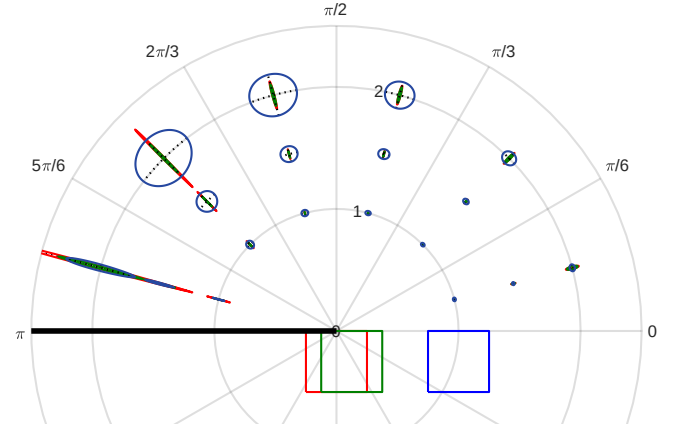
III. RESULTS

In this section, we use the tools developed in Section II to gain a deeper understanding of the active corner camera setup. We explore the effect of FOV location and motivate our choice in [cite our letter here] to model and use the occlusion provided by the vertical edge. We also study the effects of varying different experimental parameters, including FOV size, number of pixels, sensor time resolution, and laser position.

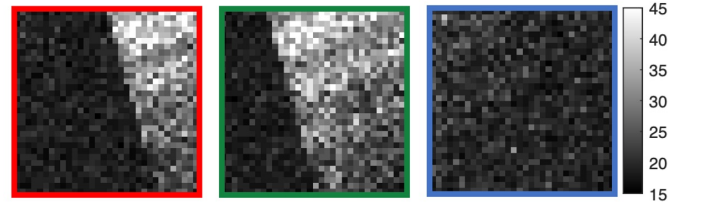
A. FOV position and occlusion

Using the CRBs derived in Section II, one can compute theoretical spatial uncertainty regions for a hidden target. These are regions within which the majority of a target's estimates are expected to fall. Specific examples for 32×32 SPAD array with an FOV of $0.5\text{m} \times 0.5\text{m}$ are shown in fig. 3a. Assuming unbiased estimators that achieve the CRB, these bubbles depict regions within which three standard deviations of a target's estimates are expected to fall when all parameters in Ψ are unknown. Bubbles are centered around the true facet location and are computed for a facet facing the occluding edge at XXm height. We observe that the uncertainty regions vary as a function of FOV position. Three different FOV locations are outlined in red, green, and blue. The red FOV location is adjacent to the occluding edge, horizontally centered so that half of the FOV extends past the origin, as in conventionally done in [cite our letter here, cite Will's paper]. The green FOV is shifted along the positive x -axis so that 75% of it extends past the origin. We selected this FOV location to have fewer pixels in shadow, and thus higher signal to noise ratio (SNR), while ensuring that a penumbra would still be observed at any angular location in the hidden scene. The blue FOV is more consistent with the position used in [1], positioned further from the vertical edge so that no pixels are in shadow for most angular locations in the hidden scene. In all three cases, the position of the sensor itself is fixed at $(x, y, z) = (0, -0.25, 1.5)$ meters. Sample measurement realizations at each FOV position, for a target at $\phi = 7\pi/12$, are shown in Figure 3b for the same laser power and background rate used to produce the uncertainty regions in Figure 3a. These measurement images were produced by taking the maximum histogram value at each spatial pixel. Counts are noticeably lower in the blue FOV measurement because that FOV is further away from the hidden target.

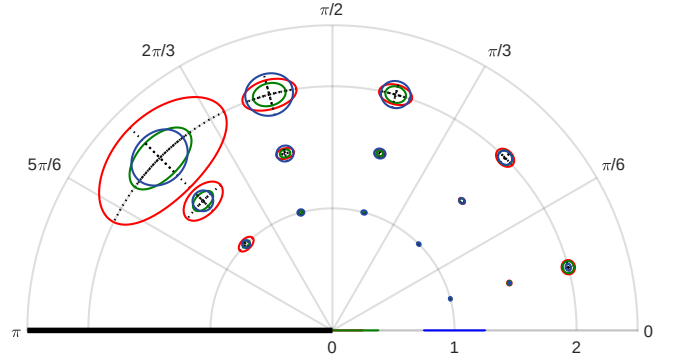
Although occlusion improves the conditioning of our inversion problem, pixels in shadow result in lower SNR. Different FOV positions are also viewed at different angles by, and are at different distances from, the sensor itself and the hidden scene, resulting in additional SNR tradeoffs. For example, the blue FOV is generally far enough along the positive x -axis to be out of the shadow cast the occluding edge but it is also further from the camera and viewed at a sharper angle. We compare the three FOV positions to better understand when



(a) Uncertainty regions for three different FOV positions when *all* camera pixels are used. The camera is centered over the red FOV at a height of 1.5m.



(b) Image of histogram max at each FOV for the target location at a range of 1m and angle of $7\pi/12$.



(c) Uncertainty regions when shadowed pixels are discarded from the measurement, as in [8]. Bubbles at larger polar angles are omitted because the uncertainty is too large to produce a clear plot.

Fig. 3: In (a), we explore the merit of different FOV locations. There are parts of the hidden space where the blue bubbles completely engulf the red. There are even more regions where the blue engulfs green. These findings support our choice to make use of the vertical edge occluder. In (b) we show uncertainty regions for the scenario where shadowed pixels are discarded from the measurement.

it is advantageous to exploit the occlusion provided by the vertical edge.

In general for all FOV positions, uncertainty grows with increasing distance from the measurement region, due to radial falloff causing reduced SNR. The blue FOV is completely illuminated (i.e. no pixels in shadow) by all angular positions except for $\phi = 11\pi/12$. For the blue FOV, these positions with no occlusion exhibit round uncertainty regions, with comparable uncertainty in range ρ and angle ϕ . For target positions at $\phi = 11\pi/12$, a penumbra shadow is cast on the blue FOV and the angular uncertainty shrinks. For the

red and green FOVs, targets at all angular positions cast a shadow on the measurement region resulting in uncertainty regions that are universally thin in angle. The green FOV, which extends further past the occluding edge so that fewer pixels are in shadow. This increase in SNR results in lower range uncertainty, with small unperceivable changes in the angular uncertainty depending on the angular location of the target. We note that, not only does measurable occlusion result in low angular uncertainty, but also that in many parts of the scene, the red and (especially) green FOVs result in lower range uncertainty as well. These findings demonstrate that it is often (i.e. for many parts of the hidden scene) advantageous to position the camera FOV adjacent to the vertical edge to exploit the occlusion it provides. Even when range uncertainty is higher, the low angular uncertainty provided by the edge occluder could be advantageous when the goal is to produce an accurate count of hidden objects.

In [8], the effects of occlusion are not modeled and the FOV is positioned away from the edge to prevent it. When occlusion does occur (for targets with larger ϕ), occluded pixels are discarded from the measurement to avoid model mismatch. In Figure 3c, we plot the uncertainty regions that arise when occluded pixels are excluded from the measurement, as they are in [8]. In this figure, uncertainty regions for targets at $\phi = 11\pi/12$ are not pictured because the corresponding uncertainty was too large to produce a meaningful plot. The blue bubbles remain the same compared to Figure 3a for the target locations pictured because, for those locations, no pixels are occluded. Because occluded pixels contain no information about target range, we note that the range uncertainty for the red and green bubbles in Figure 3c is comparable to the range uncertainty in Figure 3a. However, in Figure 3c, where the effects of occlusion are ignored, the angular uncertainty becomes much larger. At any visible side FOV location, there are certain angles ϕ within the hidden scene that will result in occlusion. Thus, regardless of the FOV location, it is advantageous to model and utilize the occlusion provided by the vertical edge, as we do in [CITE OUR LETTER].

B. FOV size

In Figure 4, we explore the effect of varying FOV size on estimate uncertainty for a single target with an albedo of 1 and at location $(\rho, \phi, h) = (1 \text{ m}, 7\pi/12 \text{ rad}, XX \text{ m})$. The SPAD FOV is $0.5 \text{ m} \times 0.5 \text{ m}$ in size and centered along the x -axis adjacent to the occluding wall like the red FOV pictured in Figure 3a. In figs. 4a to 4d, we plot $\sqrt{\text{CRB}}$ for target albedo α , range ρ , height h , and angle ϕ as a function of SPAD FOV size. Sizes tested include FOV widths of 0.25 m, 0.5 m, 1 m, and 2 m. In figs. 4e to 4h, we plot sample measurements for each FOV width, generated using the same laser intensity and background rates used to compute the Cramér–Rao bound. These images depict the maximum histogram value at each spatial pixel. Across all four FOV widths, the rate of background counts per unit area in the FOV is held constant, resulting in higher background rates per pixel as the FOV width grows. A larger FOV also results in more spatial diversity between pixels, however as

the FOV grows, certain pixels get further from the occluding edge resulting in lower signal rate per unit area. This effect is most pronounced in the large 2 m FOV in fig. 4h. We consider figs. 4a to 4d to better understand these tradeoffs.

In fig. 4a, we observe that the $1 \text{ m} \times 1 \text{ m}$ FOV results in the lowest albedo uncertainty. In figs. 4b and 4c, range and height uncertainty continue to decrease with increasing FOV size over the range tested, although the gains become diminishingly small. In general, the spatial diversity provided by a larger FOV seems to improve our ability to estimate albedo α , range ρ , and height h until the FOV becomes so large that SNR becomes harmfully low. The angular resolution provided by the vertical edge arises from the crisp boundary between occluded and unoccluded pixels; the line formed by this boundary points towards the location of the target in the hidden scene. It is clear in the sample measurements (especially fig. 4h) that as the FOV grows, this boundary becomes less pronounced. In fig. 4d we observe that, although it remains relatively small, the angular uncertainty grows with increasing FOV size.

C. Sensor pixel dimension

Although the SPAD array used in [CITE OUR LETTER] is 32×32 pixels, other existing sensors offer different pixel dimensions. In Figure 5, we explore the effect of sensor pixel dimension on estimate uncertainty. In figs. 5a to 5d we plot $\sqrt{\text{CRB}}$ for target albedo α , range ρ , height h , and angle ϕ as a function of sensor pixel dimension. Pixel dimensions tested include 8, 16, 32, and 64. Sample measurements for each case are shown in figs. 5e to 5h. As expected, the uncertainty goes down for all four unknown parameters with increasing spatial resolution. However, as shown in fig. 5d, increasing the pixel dimension decreases uncertainty of angle parameter ϕ far more than the uncertainty of other parameters. As mentioned in section III-B the angular resolution provided by the edge arises from our ability to fit a line through the boundary between illuminated pixels and those in shadow. Increasing the sensor pixel dimension makes this task much easier. However, even for the smallest pixel dimension of 8, the angular uncertainty remains low. Considering the fact that the uncertainty of the other parameters is only moderately affected by increasing pixel dimension, and that pixel density often comes at the expense of other sensor features, it might be reasonable to prioritize those other features.

D. Sensor time resolution

In Section III-E we explore the effects of varying a different sensor parameter: time resolution. Figures 6a to 6d plot the uncertainty of albedo, range, height and angle ϕ as a function of sensor time resolution. For reference, the sensor time resolution in [cite our letter!] was 390 ps. As expected, we observe that uncertainty decreases with finer time-resolution comparably for all four parameters. Thus, it may make sense to prioritize a sensor with finer time resolution over a sensor with higher pixel density.

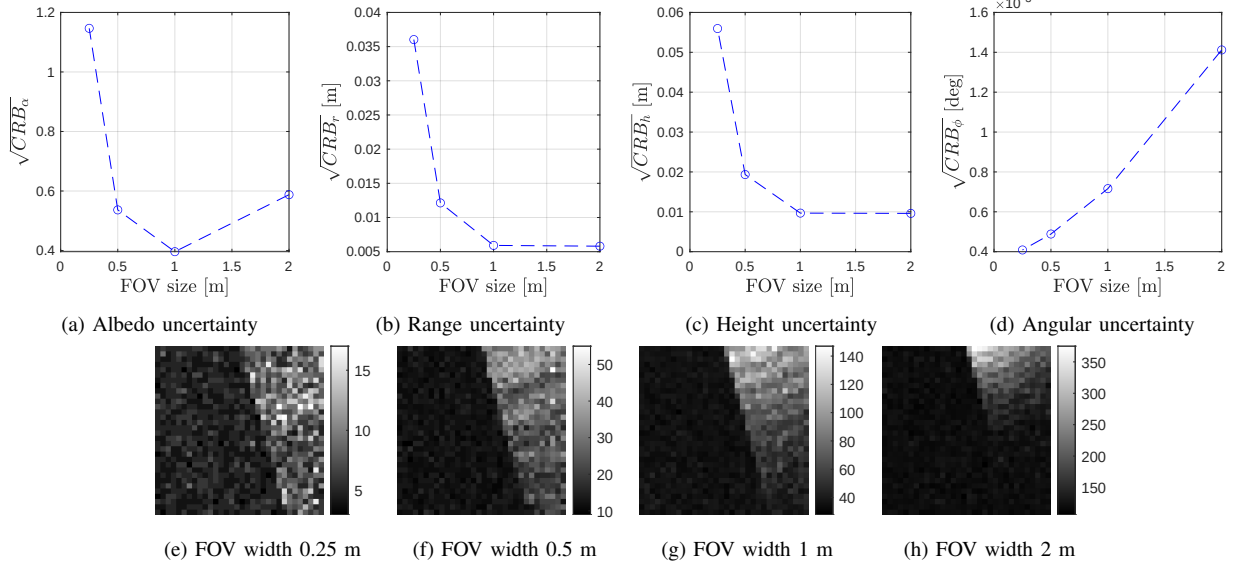


Fig. 4: The first row shows the Cramér–Rao bound for a hidden target’s albedo α , range ρ , height h , and angle ϕ , plotted as a function of FOV size in meters. The second row shows sample measurements for each FOV size, produced using the same background counts used to compute the Cramér–Rao bound. Results are for a target with albedo $\alpha = 1$ at $(\rho, \phi, h) = (1\text{m}, 7\pi/12\text{rad}, XX\text{m})$.

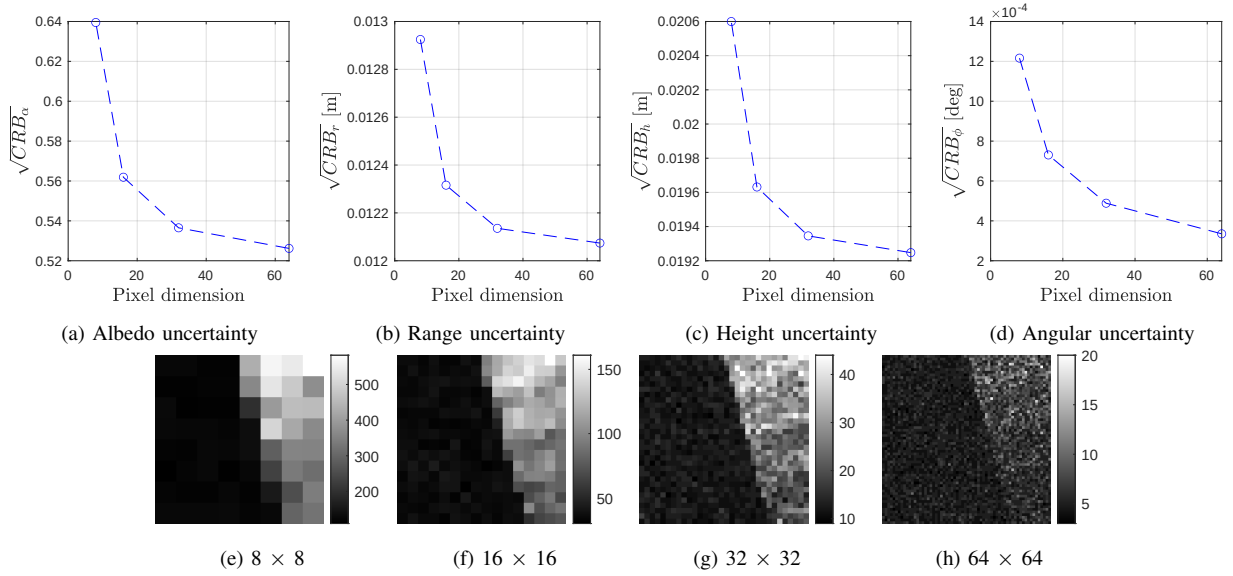


Fig. 5: The first row shows the Cramér–Rao bound for a hidden target’s albedo α , range ρ , height h , and angle ϕ , plotted as a function of pixel dimension. The second row shows sample measurements for each FOV size, produced using the same background counts used to compute the Cramér–Rao bound. Results are for a target with albedo $\alpha = 1$ at $(\rho, \phi, h) = (1\text{m}, 7\pi/12\text{rad}, XX\text{m})$.

E. Laser spot location

In [cite our letter!], the laser spot is directed at the floor and kept in a fixed position over the duration of the measurement. Figure 7 shows uncertainty regions (within which 3 standard deviations of estimates would be expected to fall) for different laser spot positions. An inset image shows a zoomed in portion of the plot, for improved readability. Bubbles are computed for a target at a height of 1 m with an 8×8 pixel sensor array with an FOV width of 0.5 m. In all three cases, the total acquisition time is the same. The red bubbles correspond to the laser fixed on the ceiling above the origin, the green bubbles correspond to the laser spot fixed on the floor at the origin,

and the blue bubbles arise when half of the acquisition time is spent in each of the two positions. For this particular target height, the ceiling position (red) results in lower uncertainty in both range and angle than the floor position (green), although we have observed this to change as a function of the target’s height. More importantly, combining measurements from two different laser positions (blue) results in comparable angular uncertainty with greatly reduced range uncertainty. In fact, the range uncertainty becomes so small that the blue bubbles collapse in range to effectively resemble a line. These results suggest that by scanning back and forth between just two laser positions, a more powerful imaging system could be realized.

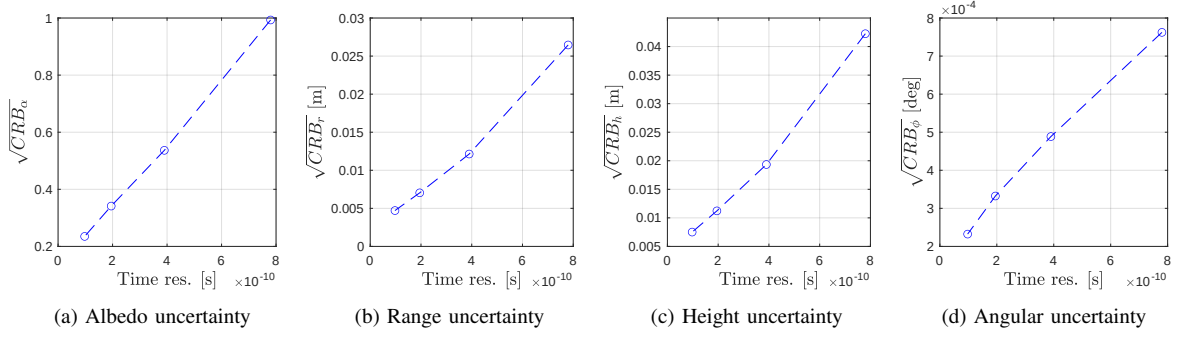


Fig. 6: The Cramér-Rao bound for each of the unknown parameters as a function of sensor time resolution for a target with albedo $\alpha = 1$ located at $(\rho, \phi, h) = (1\text{m}, \frac{7\pi}{12}\text{rad}, XX)$. The camera FOV is $0.5\text{ m} \times 0.5\text{ m}$ with a pixel dimension of 32 and the position of the sensor itself is fixed at $(x, y, z) = (0, -0.25, 1.5)\text{ m}$.

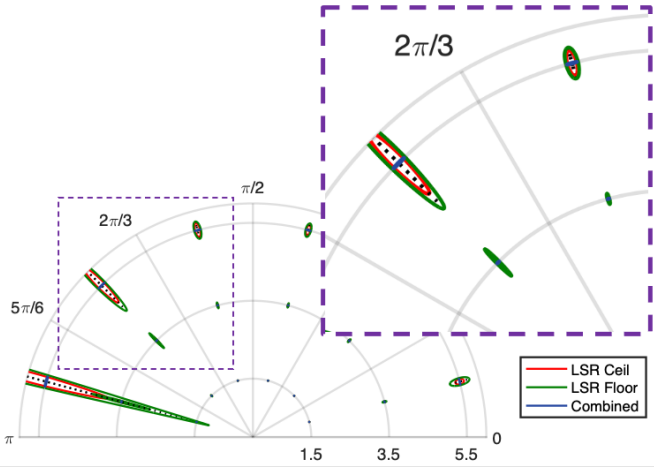


Fig. 7: Uncertainty regions for a target at 1m height for three different laser scenarios, all with the same total acquisition time. Red bubbles are for a laser spot on the ceiling, green bubbles are for a laser spot on the floor, and the blue bubbles arise when the acquisition time is divided between the floor and ceiling. An 8×8 pixel array was used to increase angular uncertainty for better visibility.

IV. CONCLUSION

In this work, we derive the Cramér-Rao bound for estimating a small hidden facet's unknown parameters using measurements from the active corner camera. Using our Cramér-Rao bound tools, we show the merit of exploiting the occlusion provided by the vertical edge [cite our letter] over the approach of avoiding it [8]. Although previous corner camera works have centered the camera FOV around the occluding edge, our findings suggest that, for many parts of the hidden scene, even lower uncertainty can be achieved by slightly shifting the FOV to expose more pixels to the hidden scene. We expect this finding to extend to the passive corner camera as well. Our plots of uncertainty versus various parameters could be used to optimize the active corner camera system under different operational constraints.

ACKNOWLEDGMENT

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